

Proofs by Substitution

Lecture Notes

Course: Computer-Assisted Theorem Proving (7CCSACTP)

Context & Concept of Substitution

→ **Previous Videos:**

- Introduced Isabelle interface
- How to write terms / theorems in Isabelle
- Two basic proof / reasoning approaches:
 - Forward reasoning (FW)
 - Backward reasoning (BW)

→ **This Video: Proofs by Substitution**

- Suppose we have a theorem:

lem1: $\{?x+?y=?y+?x\} \longrightarrow \text{simplication rule}$

- And we have a goal: **goal** :: $1 + 3 > 3 + 1$
- Substituting **lem1** into the goal gives a new goal:

goal :: $3 + 1 > 3 + 1$

- **When does substitution make sense?**

- If R.H.S. of the simp rule is simpler than L.H.S., or if we have a proof on R.H.S.

Worked Example: Distributivity on Nats

Goal: $\vdash (x + y) \cdot (a + b) = x \cdot a + x \cdot b + y \cdot a + y \cdot b$ (Note: all variables are nat)

- $(x + y) \cdot (a + b) = (x + y) \cdot a + (x + y) \cdot b$
→ Using rule: $?x \cdot (?a + ?b) = ?x \cdot ?a + ?x \cdot ?b$ (**lem1**)
- $\underline{(x + y) \cdot a} + (x + y) \cdot b = x \cdot a + y \cdot a + (x + y) \cdot b$
→ Using rule: $(?a + ?b) \cdot ?x = ?a \cdot ?x + ?b \cdot ?x$ (**lem2**)
- $\dots = x \cdot a + y \cdot a + x \cdot b + y \cdot b$ ✓
→ Using distributive and associative lemmas.

Isabelle Practical Notes

- **Show proof steps:** How to perform the proof step-by-step using rules.
- **Library Search:** Find lemmas that already exist in Isabelle's library.
- **Reflexivity:** A goal of the form $x = x$ can be proved using `by (rule refl)` or `by simp`.
- **Associativity convention:** $a + b + c$ is parsed as $a + (b + c)$.
- **Symmetric attribute:** If `lem1: x = y`, then `lem1[symmetric]` gives $y = x$.
- **Closing proofs:** If the goal is $P \vdash P$, close it by `."` (or by `assumption`).
- **Calculational Reasoning (Isar):**

```
have          "x = y"      ...
also have     "... = z"    ...
also have     "... = a"    ...
also have     "... = b"    ...
finally show  "x = b"      .
```